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METHOD OF STATEMENT FOR PRESSURE TESTING OF COPPER PIPE (REFRIGERANT)

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1.0 Objective

The primary objective of this method statement is to provide a systematic and controlled procedure for conducting a pressure leak test on the entire VRF refrigerant copper piping network. This test is designed to verify the integrity and leak-proof nature of all joints and connections before the system is evacuated and charged with refrigerant.

2.0 Purpose

The purpose of this procedure is to ensure that the VRF refrigerant copper piping system is completely sealed and free from any leaks before the system is evacuated and charged with refrigerant. Integrate WSH best practices to eliminate or mitigate all identified hazards associated with high-pressure gas testing.

This process is vital for the long-term reliability and performance of the ACMV system. Serve as the primary reference for formal inspection, witnessing, and obtaining official sign-off from the Client and RTO / Maincon M&E.

3.0 Scope of Work

This procedure applies to all newly installed field refrigerant piping for the VRF system.

Inclusions:

- All copper refrigerant piping, including the main liquid and gas lines.
- All branch lines connecting the main pipes to individual Fan Coil Units (FCUs) / Indoor Units (IDUs).
- All field-fabricated brazed joints, fittings, and flare connections.
- The scope extends from the isolation valves on the Outdoor Condensing Units (ODUs) to the isolation valves on the Indoor Units (IDUs).

Exclusions from Test:

- The factory-sealed Outdoor and Indoor units themselves, which are tested by the manufacturer. The test is conducted by closing the service valves on these units, isolating the newly installed pipework.

4.0 Reference:

- Air-Conditioning and Mechanical Ventilation in Buildings. This is general guidance on the design, installation, testing, and commissioning of ACMV systems
- **VRF System Manufacturer's Installation Manual:** [e.g., Daikin, Mitsubishi Electric, etc.]
- **Project-Specific Specifications:** All requirements outlined in the project's technical specifications.
- **Workplace Safety and Health (WSH) Act** and associated regulations.
- Approved Project Specifications & Refrigerant Piping Schematics and Plan Drawings.

5.0 Responsibility

Role	Key Responsibilities
Project Manager	Ensures overall project compliance, resource availability (especially certified personnel and calibrated equipment), and client liaison.
Site Coordinator/ Supervisor	On-site management and supervision of the entire test. Ensures strict adherence to this MS and the manufacturer's procedures. Manages the PTW, records all data, and coordinates witness inspections.
WSH Supervisor	Responsible for witnessing the test, verifying the calibration of test equipment, inspecting all joints, and formally signing off on the test reports.
Confidence Technician	Enforces compliance with the approved Risk Assessment (RA). Conducts pre-work toolbox talks, ensures the test area is correctly barricaded, and monitors for unsafe acts, with full authority to stop work if necessary.
	Executes the physical work under direct supervision. Must be trained for high-pressure copper brazing. Responsible for system pressurization, leak checking, and depressurization.

6.0 Preparation (Pre-Test Readiness)

Prior to commencing the pressure test, the following preparatory steps must be completed:

1. **Documentation:** An approved **Permit-to-Work (PTW)** for pressure testing is mandatory. The relevant refrigerant piping schematic must be available, marked-up to show the section under test.
2. **System Readiness:** All piping installation and support works must be 100% complete. All brazing must be completed by confidence personnel.
3. **Isolation:** Confirm that the service valves on all Outdoor and Indoor units connected to the piping network are securely closed, isolating the pipework for the test.
4. **Test Gas:** Ensure a sufficient supply of **Oxygen-Free Dry Nitrogen (OFDN)** is on site.
5. **Equipment Setup:** Connect the nitrogen cylinder via the regulator and gauge manifold to a service port on the piping system (typically on the liquid or gas line).
6. **Pressure Regulation:** A **multi-stage nitrogen pressure regulator** in good working condition is mandatory. This is a critical safety device to prevent over-pressurization.
7. **Calibration:** All pressure gauges must have a valid calibration certificate from a **accredited laboratory**. The gauge range must be appropriate for the high-test pressure.

7.0 Hand Tools & Equipment

Category	Item & Specification
Primary Test Equipment	1. Oxygen-Free Dry Nitrogen (OFDN) Cylinder. 2. Multi-Stage Nitrogen Pressure Regulator 3. Calibrated Pressure Gauge Manifold Set (e.g., R410A/R32 rated). 4. High-pressure rated charging hoses.
Leak Detection	1. Leak Detection Soap Bubble Solution (commercial grade).
Brazing (for repairs)	1. Turbo Torch with an Oxygen/Acetylene set. 2. Brazing rods with appropriate silver content
General Hand Tools	1. Torque wrench and flare spanners for flare connections. 2. Tube cutters and deburring tools.
Safety Equipment	1. Safety Hard Barriers. 2. Warning Signs: "DANGER - HIGH-PRESSURE GAS TEST IN PROGRESS - KEEP OUT".
PPE	1. Safety Helmet, Safety Goggles, Safety Boots, Hand Gloves.



Turbo Torch with an Oxygen/Acetylene set



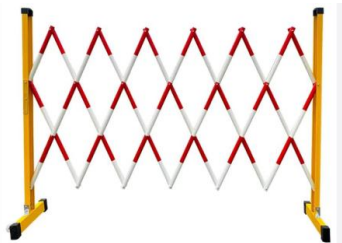
Brazing rods with appropriate silver content



Tube cutters and deburring tools



Warning Signs



Barriers.

Safety



Torque Wrench with socket set.

 Calibrated Pressure Gauges 800psi	 Pressure Gauge Manifold 600psi	 Leak Detection Solution (Soap bubble solution)
 PPE	 Ladder	 Spanner tool

8.0 Testing Method of Statement and Procedures

Test Pressure

The test pressure will be in accordance with the VRF manufacturers. For modern systems using R410A or R32 refrigerant, this is typically:

- **Test Pressure: 380~400 PSI**

8.1 Two-Stage Testing Procedure

To ensure safety and effectiveness, the test is conducted in two distinct stages.

Stage 1: Initial Low-Pressure Leak Check

1. Connect the OFDN cylinder via the regulator and gauge manifold to the system.
2. **Intermediate Pressurization:** Gradually increase the pressure. A common practice is to hold at 200 PSI for 30 minutes, then increase to 300 PSI for 1 hours
3. Meticulously apply the soap bubble solution to **every single brazed joint, flare nut, and threaded connection** in the entire network.
4. If any leak (indicated by growing bubbles) is found, the joint will be marked. The system will be fully depressurized to 0 PSI, the joint will be re-brazed by a certified technician, and Stage 1 will be repeated until all joints pass.

Stage 2: Full Pressure Standing Test

1. Once Stage 1 is successfully completed, continue to slowly introduce until the system reaches the full test pressure of **(380- 400 PSI)**.
2. Close the valve on the nitrogen cylinder and isolate it from the system.

3. **Formal Test Start:** Allow the pressure to stabilize for 30 minutes. The start time, initial test pressure, and the ambient temperature shall be recorded and witnessed by the FM/RTO or Consultant.
4. **24-Hour Holding Period:** The system shall hold the full test pressure for a minimum duration of **24 hours**.
5. **Test Conclusion**
After 24 hours, the final pressure and ambient temperature will be recorded.
Acceptance Criteria: The test is **PASSED** only if the pressure reading, after being corrected for any change in ambient temperature, has **not dropped**. Any pressure drop not attributable to a decrease constitutes a test failure. The test must be repeated after finding and rectifying the leak.

8.2 Procedure in Event of Failure

1. If a leak is detected at any stage, or if the pressure drops below the accepted value, the test has failed.
2. The location of the leak(s) must be clearly marked.
3. The system must be safely depressurized completely.
4. The faulty joint(s) must be repaired by a certified technician. This may involve re-brazing or re-flaring the connection.
5. After the repair is completed and visually inspected, the entire pressure testing procedure (starting from Section 8) must be repeated.

8.3 Post-Test Procedure

Upon successful completion and sign-off, the pressure will be safely and slowly vented to the atmosphere, and the system will be ready for the triple evacuation (vacuuming) procedure.

1. **Evacuation (Vacuuming):** The system must be evacuated using a vacuum pump to remove all non-condensable gases and moisture. A deep vacuum of 500 microns or lower should be achieved and held to confirm the system is dry and leak-free.
2. **Refrigerant Charging:** The correct amount of refrigerant, as specified by the manufacturer, is then charged into the system.
3. **System Start-up:** The system is then started, and operational parameters such as suction and discharge pressures, and compressor current draw are checked against the manufacturer's specifications.
4. **Functional Tests:** All indoor units are tested in both heating and cooling modes to verify proper operation. A full test and balance (TAB) report for each fan coil may be completed.

9.0 Documents Handover

Upon successful completion, a formal **Refrigerant Piping Test Pack** will be compiled and submitted for each tested system.

Document Control Cover Sheet.

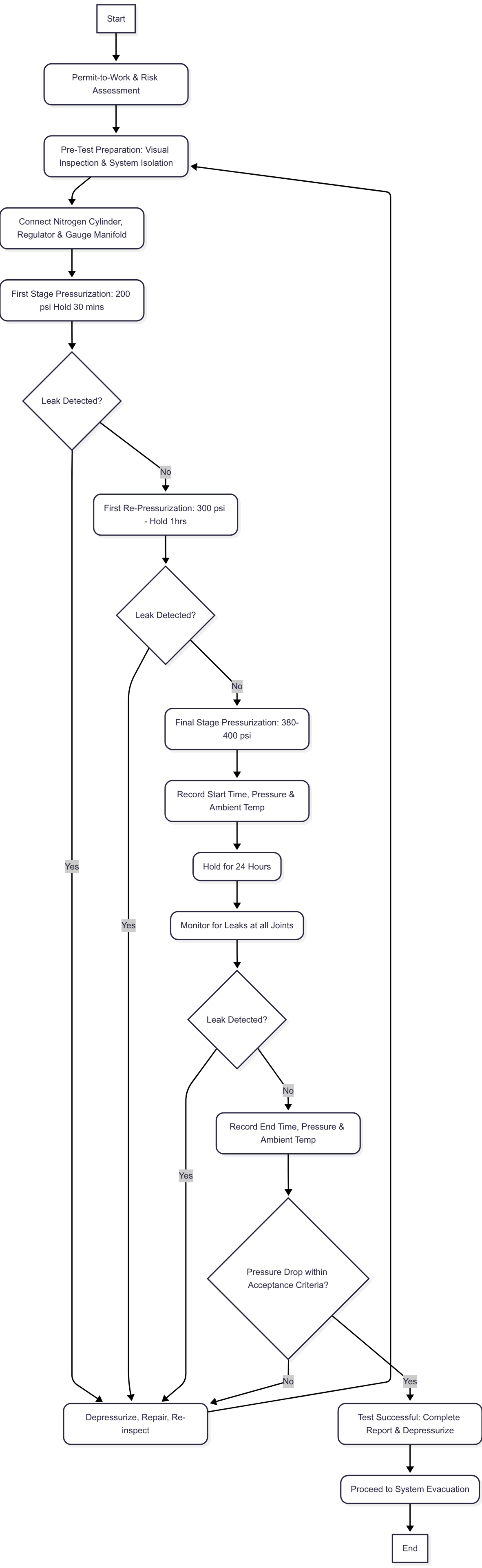
1. **Completed & Signed Pressure Test Certificate**, detailing location, test pressure, Start Date /Time and End Date /Time, duration, and final result. It must be signed by the Main Con M&E, RTO Inspector, and the Client's Representative.

2. **The Marked-up "As-Built" Drawing** clearly highlighting the successfully tested section.
3. Copies of the **Calibration Certificates** for the specific pressure gauges used.
4. The final, closed-out **Permit-to-Work (PTW)**.

10.0 Safety & Health

Safety is paramount during the pressure testing process. The following safety and health measures must be strictly adhered to:

- **Risk Assessment:** A site-specific risk assessment shall be conducted before commencing any work.
- **Personal Protective Equipment (PPE):** All personnel involved must wear appropriate PPE, including safety glasses/goggles and gloves.
- **Handling of Nitrogen:** Nitrogen cylinders must be handled with care and secured in an upright position. The pressure regulator must be used to control the pressure.
- **Warning Signs:** The work area must be clearly marked with warning signs to indicate that pressure testing is in progress.
- **Ventilation:** Ensure adequate ventilation in the work area, especially if a refrigerant leak is ever suspected, as some refrigerants can produce toxic gas if they come into contact with a flame.
- **Trained Personnel:** Only trained and competent technicians shall be permitted to carry out the pressure test.
- **Emergency Procedures:** All personnel should be aware of the emergency procedures in case of an accident.
- **Flammable Refrigerants:** If using a flammable refrigerant like R32, strict ignition control measures must be in place, and all potential ignition sources must be eliminated from the vicinity. Service personnel must be specifically trained for handling such refrigerants



**PRESSURE TEST REPORT**

PROJECT NAME			
LOCATION		EQUIPMENT DESIGNATION	
AREA SERVED		EQUIPMENT MODEL	
SYSTEM		MANUFACTURER	

PIPE SIZE	
TYPE OF GAS / WATER	
INITIAL HOLDING PRESSURE (PSI/bar)	
TEST START ON DATE / TIME	
FINAL PRESSURE READING (PSI/bar)	
DURATION OF TEST (HOURS)	
TEST END ON DATE / TIME	

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			



GRACECAL PTE LTD

(GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CLIENT : **WINNER ENGINEERING PTE LTD**
18 Woodlands Industrial Park E1
Singapore 737738

CERTIFICATE NO : GCP241096
ISSUE DATE : 11-JUNE-2024

WORKS ORDER NO : GWO241359
DATE RECEIVED : 07-JUNE-2024

INSTRUMENT : **COMPOUND GAUGE**
MAKE : BLACK DIAMOND
MODEL NO. : -----
PART NO. : -----
SERIAL NO. : -----

DATE CALIBRATED : 11-JUNE-2024
DUE DATE : 11-JUNE-2025

AMBIENT TEMPERATURE:
(23 ± 5) °C
RELATIVE HUMIDITY:
(55 ± 10) % r.h.

RANGE : -30 inhg~800 psi
GRADUATION : -----
(TAG NO.) : PG11/R3

PAGE : 1 of 2
STATUS : AS FOUND

The described instrument has been calibrated at Laboratory under the ambient conditions stated above.

The reference measurement standards used are traceable to National Metrology Centre (**NMC**), Singapore or National Physical Laboratory (**NPL**), UK or National Institute of Standards and Technology (**NIST**), USA or other recognized National or International Standards Laboratories.

The Instrument was calibrated by comparison with laboratory calibration testing equipment.

Equipment: Pressure Calibrator
Serial No: 3866884

Calibration Method

Calibration Method is in accordance with **ISO/IEC 17025**. Calibration Method was carried out according to in-house Technical Calibration Procedure AS-TM-M01 as a guide.

RESULTS OF CALIBRATION

1. The results of calibration are given on the attached calibration data sheet(s).
2. The expanded uncertainty of measurement associated with the calibration estimated at a confidence level of approximately 95% with a coverage factor of k=2.00.
3. The user should determine the suitability of the instrument for its intended use.

Calibrated by :

PREM
ID: 748656

Approved by :

KUMAR
ID: 748670





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CALIBRATION CERTIFICATE

CERTIFICATE NO : GCP241096
ISSUE DATE : 11-JUNE-2024

WORKS ORDER NO : GWO2412359
PAGE : 2 of 2

APPLIED ACTUAL VALUE (PSI)

Ascending

Error

Before Adj.

After Adj.

MEAN INSTRUMENT READING (PSI)

Descending

Error

Before Adj.

After Adj.

VACUUM MEASUREMENT: (inHg)

0.0	0	---	0.0	0	---	0.0
-30.0	-30	---	0.0	-30	---	0.0

PRESSURE MEASUREMENT: (psi)

0.0	0	---	0.0	0	---	0.0
196.4	200	---	-3.6	200	---	-3.6
296.4	300	---	-3.6	300	---	-3.6
596.4	600	---	-3.6	600	---	-3.6
796.4	800	---	-3.6	800	---	-3.6